Monocular Metric Depth with Video Consistency

Anonymous CVPR submission

Paper ID *****

Abstract

We study depth from a single RGB image and from a short monocular video, in three steps. First, StereoSGBM on the 15 Middlebury quarter-resolution training pairs turns disparity into meters with the supplied calibration, with a mean absolute disparity error of 1.42 px on the 62% of pixels it keeps. Second, UniDepthV2-Small predicts metric depth on the official 654-image NYU Depth v2 test split from RGB alone, with no test-time alignment (AbsRel 0.115, δ_1 0.892). On the provided `davis_rollercoaster` *clip we compare frame-wise Depth Anything V2-Small with Video Depth Anything-Small under one fixed normalization per method. Third, we fit one global scale on the 795 NYU training images and freeze it for the test split. That single number, $s = 0.956$, lowers test AbsRel to 0.102 and raises δ_1 to 0.900, and it is stable from about 50 training images onward. A depth-aware background blur is the application demo. Code:* https://github.com/Oliver-ming/Intro_to_CV_project_3.

1. Introduction

A single photograph does not determine metric depth. The same image is explained by a small nearby scene or a large distant one. Stereo removes that ambiguity with a known baseline, and depth then follows from disparity. Monocular networks remove it only by learning what scenes usually look like, so a model can rank pixels well and still be wrong in meters.

We keep three kinds of output apart throughout. *Stereo depth* comes from disparity and calibration. *Relative depth*, from Depth Anything V2 [13] or Video Depth Anything [2], is defined only up to an unknown scale and shift, and we never put it in meters. *Metric depth*, from UniDepthV2 [6], is scored on NYU with no test-set fitting. Video adds a second problem: a frame-wise model can change its answer between frames, which shows up as flicker. The provided clip has no ground truth, so that part is qualitative.

For the improvement we pick metric scale calibration. UniDepthV2-Small, run RGB-only on NYU, overestimates depth by a few percent on most images. If the bias is systematic for a domain, one number estimated on training data should remove part of it without touching test labels. We measure how much, with which estimator, from how few images, and where it fails. Figure 1 shows the three parts.

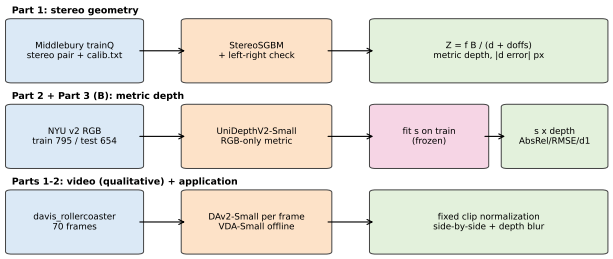


Figure 1. Pipeline. Top: stereo geometry. Middle: metric depth and the frozen global scale (Part 3, direction B). Bottom: the video comparison and the depth-aware blur demo.

2. Related Work

Stereo. Semi-global matching aggregates matching costs along several 1-D paths to approximate a 2-D smoothness prior [4]; OpenCV's StereoSGBM is a block-matching variant of it. KITTI made stereo and laser depth a standard driving benchmark [3]. We use Middlebury's quarter-resolution training pairs, which come with dense ground truth and per-scene calibration [8].

Monocular foundations. DPT replaced convolutional depth decoders with a vision-transformer backbone and a reassembling head [7]. Depth Anything scales that design with tens of millions of pseudo-labeled images [12]. Depth Anything V2 trains its teacher on synthetic images and distills it to smaller students, which gives sharper boundaries [13]. Its released Small checkpoint predicts relative depth.

Metric depth. UniDepth predicts depth and a dense camera representation together, so it needs no intrinsics at test

time [5]. UniDepthV2 simplifies that design, adds an edge-guided loss and an uncertainty output, and still accepts a known camera as an optional input [6]. Depth Pro estimates focal length and produces sharp metric depth in under a second [1]. These models are zero-shot: none is trained on our NYU split. That leaves room for a domain bias that a single scale could absorb.

Video depth and motion. RAFT computes dense optical flow by iterative updates on an all-pairs correlation volume [11]; we use it only for an optional temporal diagnostic. Video Depth Anything adds a spatio-temporal head to Depth Anything V2 and trains it with a temporal gradient-matching loss, then stitches long videos from overlapping 32-frame windows [2]. NYU Depth v2 provides the labeled indoor images and the official split [9]. TUM RGB-D is a video benchmark with ground-truth depth and poses [10]; the course does not require it and we do not use it.

3. Method

3.1. Part 1: stereo geometry

On each rectified Middlebury pair we run StereoSGBM (3-way mode, block size 5, $P_1 = 8 \cdot 25$, $P_2 = 32 \cdot 25$, uniqueness 10, speckle window 100) with as many disparity levels as `ndisp` in `calib.txt`, rounded up to a multiple of 16. OpenCV returns fixed-point disparity, which we divide by 16. We then match right-to-left and keep a left pixel only if the two disparities agree within 1 px. A prediction is *invalid* if SGBM finds no match ($d \le 0$) or the pixel fails this check. Valid pixels become meters with the Middlebury calibration,

$$Z = \frac{f\,B}{d + d_{\text{offs}}} \cdot \frac{1}{1000}, \tag{1}$$

where f is the quarter-resolution focal length in pixels, B the baseline in millimeters, and d_{offs} the principal-point offset between the two cameras. Differentiating gives $|\partial Z/\partial d| = 1000\,Z^2/(fB)$, so the same disparity error costs far more depth at long range.

3.2. Part 1–2: frame-wise and video relative depth

Depth Anything V2-Small (DAv2-S) runs independently on every frame of `davis_rollercoaster.mp4` at the repository's default input size 518. Video Depth Anything-Small (VDA-S) runs offline on the same frames with the official flags `--encoder vits --max_len 100 --max_res 640` and FP16. The clip has only 70 frames, so "the first 100 frames" is the whole clip, read at its original 24 fps and resized to 640×360. Both outputs are relative disparity-like maps. We color each method with one fixed range over the whole clip (its 2nd and 98th percentiles) and the same palette. Ranges differ between the two methods, because their outputs share no scale. We never fit these maps to meters.

3.3. Part 2: zero-shot metric depth

UniDepthV2-Small (ViT-S, `lpiccinelli/unidepth-v2-vits14`) receives only the RGB image through `infer(rgb)` and predicts its own camera and metric depth. We resize its output to 480×640 bilinearly. No scale or shift is fit on the test split. As an ablation we also pass the published Kinect intrinsics, which the brief does not require.

3.4. Part 3: frozen global metric scale (direction B)

Let D_i be the raw UniDepth prediction and G_i the ground truth of training image i, and let Ω_i be its evaluation pixels (same crop and range as the test protocol). We fit one scalar s on the 795 training images only and report test metrics of $s\,D$. The main estimator gives each image one vote,

$$s_{\text{med}} = \underset{i \in \text{train}}{\text{median}}\ \underset{p \in \Omega_i}{\text{median}}\ \frac{G_i(p)}{D_i(p)}, \tag{2}$$

so a few images with very large or very small depth cannot pull the estimate. For the ablation we pool all training pixels and minimize three losses: squared log error, $s_{\log} = \exp(\overline{\log G - \log D})$; squared metric error, $s_{\text{ls}} = \sum DG / \sum D^2$; and AbsRel itself. Because $\sum |sD - G|/G = \sum (D/G)\,|s - G/D|$, the AbsRel minimizer is the weighted median of G/D with weights D/G. The pooled estimators weight images by their number of valid pixels.

We choose s_{med} before looking at test results, because its per-image vote matches the test protocol, which averages metrics per image. Once fit, s is written to disk and never changed. A global scale cannot change depth ordering or the relative shape inside an image. It can fix only a bias that is shared across images, which sets an upper limit on what this direction can achieve. We report that limit as a diagnostic *oracle*: a per-image median scale fitted to test ground truth, which is not a valid method.

3.5. Application: depth-aware background blur

Given a clip-normalized disparity map $\hat{d} \in [0, 1]$ (1 is near) and a focus level $\phi = 0.55$, the blur amount at a pixel is $a = \text{clip}((\phi - \hat{d})/\phi, 0, 1)$. Pixels at or in front of the focus plane stay sharp; farther pixels are blended between six Gaussian levels up to $\sigma = 9$ px. The same ϕ and clip normalization drive the effect for DAv2-S and VDA-S, so any difference in the output comes from the depth alone.

4. Experiments

4.1. Protocol and hardware

NYU. Official `splits.mat`: `trainNdxs` (795) for fitting, `testNdxs` (654) for evaluation,

Scene	$\lvert d$ error$\rvert$ (px)	valid fraction
Teddy / Vintage / Recycle	0.48 / 0.62 / 0.66	0.73 / 0.48 / 0.73
Adirondack / Motorcycle	0.85 / 1.00	0.73 / 0.76
Pipes / ArtL	1.65 / 1.59	0.68 / 0.46
PianoL / Jadeplant	4.08 / 4.17	0.40 / 0.38
Mean of 15 pairs	**1.42**	**0.622**

Table 1. StereoSGBM on Middlebury trainQ. Error is the mean absolute disparity error over valid predictions on finite, positive ground truth. Invalid: no SGBM match or a left-right disagreement above 1 px. Per-scene values are in `outputs/stereo`.

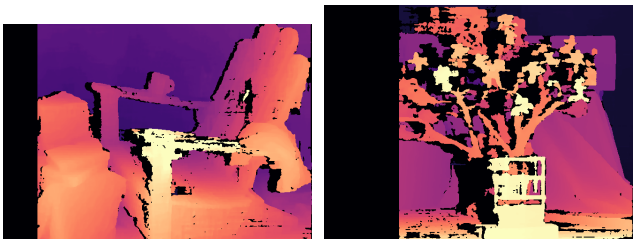


Figure 2. SGBM disparity, one fixed range per image. Left: Adirondack, one of the easier scenes. Right: Jadeplant, where thin leaves and occlusions leave large holes.

with MATLAB indices shifted by one. RGB is `images[i].transpose(2,1,0)` and ground truth is the filled `depths[i].T` in meters. Predictions are resized to 480×640 and clipped to $[0.001, 10]$ m. We evaluate the crop `[45:471, 41:601]` where ground truth is in $(0.001, 10)$ m and average per-image AbsRel, RMSE, and δ_1 equally over the 654 images. These are our own runs under the course protocol, not a reproduction of a published number. **Stereo.** The 15 `trainingQ` pairs; we score on finite, positive `disp0GT.pfm` pixels. **Hardware.** One NVIDIA RTX 4090 (24 GB), PyTorch 2.11, CUDA 12.8. Runtime excludes data loading and is averaged after warm-up; peak memory is `torch.cuda.max_memory_allocated`.

4.2. Stereo

Table 1 reports the course metrics. SGBM keeps 62% of ground-truth pixels, and on those its mean absolute disparity error is 1.42 px. Repetitive or thin structures (Jadeplant's leaves, PianoL's lighting change) are the worst. In Adirondack ($f = 1038$ px at quarter resolution, $B = 176$ mm), Eq. 1 turns a 1 px error into 5.5 mm at 1 m but 350 mm at 8 m.

4.3. NYU metric depth and the frozen scale

Table 2 is the main result. RGB-only UniDepthV2-S reaches AbsRel 0.1145. On training images the per-image ratio $\mathrm{median}(G/D)$ has median 0.956 and interquartile range 0.905–1.008, so the model overestimates NYU depth by about 4% on a typical image. The frozen scale $s_{\mathrm{med}} = 0.9563$ lowers test AbsRel by 11% relative, to 0.1020, and also lowers RMSE and raises δ_1. All three metrics move in the same direction, so the gain is not an artifact of optimizing AbsRel. The scale helps 62.8% of the 654 test images and hurts the rest.

Method	s	AbsRel↓	RMSE↓	δ_1 ↑
UniDepthV2-S, raw	1	0.1145	0.399	0.892
+ frozen s_{med} (ours)	0.956	**0.1020**	**0.389**	**0.900**
Diagnostic, uses test GT, not a method				
Per-image median oracle	–	0.0604	0.295	0.950

Table 2. NYU Depth v2 official test split (654 images), RGB-only inference, course protocol. s is fit on the 795 training images only.

Camera	Estimator	s	AbsRel	RMSE	δ_1
RGB-only	raw	1	0.1145	0.399	0.892
	s_{med}	0.956	0.1020	0.389	0.900
	$s_{\log}$	0.955	0.1019	0.390	0.899
	s_{ls}	0.937	0.1017	0.397	0.896
	s_{AbsRel}	0.943	0.1015	0.394	0.898
Kinect K	raw	1	0.0922	0.358	0.926
	s_{med}	0.989	0.0908	0.359	0.929
	s_{ls}	0.951	0.0945	0.382	0.928

Table 3. Scale estimator and camera ablation. Every s is fit on NYU train and frozen for test.

The oracle row shows how far one number per image could go (AbsRel 0.060). One number for the whole dataset closes 23% of the gap between raw and oracle. The remainder is image-specific scale error and shape error, which a global scale cannot reach.

Ablation: estimator. Table 3 (top) refits s on the same training images with each loss. All four estimates lie in 0.937–0.956 and give test AbsRel within 0.0005 of each other. The pooled least-squares fit gives the smallest s, because squared metric error is dominated by far pixels, and it is the worst on RMSE and δ_1. The choice of estimator matters much less than whether a bias correction is applied at all.

Ablation: known camera. Passing the Kinect intrinsics (Table 3, bottom) already removes most of the bias: raw AbsRel drops to 0.0922 and the fitted scale moves to 0.989. The frozen scale then gains only 0.0014 AbsRel, and the least-squares variant makes AbsRel worse. This identifies the source of the RGB-only bias. UniDepth's predicted focal length, not its depth shape, carries most of the systematic NYU scale error, and the frozen scale corrects it on average when no camera is given.

Does it generalize? We refit s_{med} on random training subsets, 20 draws per size (Table 4). With 5 images the

Train images	s (mean $\pm$ std)	Test AbsRel (mean $\pm$ std)
5	0.962 ± 0.036	0.1078 ± 0.0113
20	0.960 ± 0.020	0.1041 ± 0.0026
50	0.961 ± 0.014	0.1035 ± 0.0021
200	0.958 ± 0.005	0.1024 ± 0.0006
795	0.956	0.1020

Table 4. Fitting s_{med} on random subsets of NYU train (20 draws per size, seed 0). Raw AbsRel is 0.1145.

Scene type	#test	#train	raw	$+s$
bedroom	191	192	0.104	**0.089**
living room	107	114	0.117	**0.112**
kitchen	106	119	0.117	**0.105**
bathroom	58	63	0.144	**0.134**
dining room	55	62	0.102	**0.090**
office	38	40	0.113	**0.087**
bookstore	11	25	0.190	**0.180**
reception room	5	12	**0.112**	0.138

Table 5. Test AbsRel per NYU scene type (selected rows; all 16 types are in `per_scene.csv`).

scale has a standard deviation of 0.036 and the test AbsRel varies by 0.011. From 50 images onward the scale is within 0.014 of the full fit and test AbsRel is within 0.0013, so the correction does not depend on any particular training images. Per scene type (Table 5), the frozen scale lowers AbsRel in 13 of 16 types, including all eight with at least 20 test images. It hurts three small types (reception room, computer lab, foyer; 10 test images in total), too few to tell a different bias from noise. The improvement therefore transfers across NYU's indoor scene types, but it is a property of this domain. A different camera or an outdoor dataset would have a different bias, and we would not expect 0.956 to transfer.

4.4. Failure cases

We take the three test images with the highest raw AbsRel (Fig. 3, rows 2–4) and sort them by the per-image median ratio $r = \text{median}(G/D)$. **Global scale failure.** In the study ($r = 0.75$, AbsRel 0.563) and the bookstore ($r = 0.75, 0.398$), the whole scene is predicted about 33% too far. The frozen scale moves both in the right direction (to 0.500 and 0.360), but 0.956 is far from 0.75. Only a per-image cue such as a known camera or an object of known size could fix these. **Shape failure.** In the bedroom ($r = 0.93$, AbsRel 0.381) the median scale is nearly right, yet the error stays high. The error is in the layout of the room, so no scale can repair it. **Scale hurts.** The living room in row 5 ($r = 1.075$) was already predicted slightly too near, and the scale raises its AbsRel from 0.066 to 0.107. A global correction always moves some images the wrong way; Table 2 reports the net effect.

Model	Input	Precision	ms/frame	Peak MiB
UniDepthV2-S (NYU)	480×640	FP32	16.2	432
DAv2-S (clip)	518 short side	FP32	36.3	415
VDA-S (clip, offline)	518 short side	FP16	60.9	8016
SGBM + LR check	trainQ	CPU	–	–

Table 6. Runtime and peak GPU memory on one RTX 4090. VDA-S processes 32-frame windows, which sets its memory. The frozen scale is one multiplication and adds no measurable cost.

4.5. Video: frame-wise versus video depth

Figure 4 shows the same frames from both methods, each under its own fixed clip range. Both recover the track, the cars, and the sky as the farthest region. The comparison is qualitative, because the clip has no ground truth; the numbers below are an optional diagnostic and say nothing about metric accuracy. As a diagnostic we warp each clip-normalized map onto the next frame with RAFT-Small and average the absolute change on pixels whose forward and backward flow agree within 1 px. VDA-S changes less than DAv2-S on all 69 frame pairs (mean 0.0143 versus 0.0170 of the clip range, a 16% reduction). The per-pixel map of this change (Fig. 5) shows where the frame-wise model flickers. A remaining source of change in both maps is real motion that the flow does not fully explain, such as occlusion boundaries around the moving cars. **[TODO: Confirm the visual description of Fig. 4 and Fig. 5 against the videos in `outputs/video` before submission.]**

4.6. Application: depth-aware blur

Figure 6 applies the same blur rule to both depth maps. The near track and cars stay sharp and the background is blurred. Because the rule reads clip-normalized depth with one fixed focus level, a frame-wise model that changes its depth from frame to frame also changes which pixels are blurred. The blur videos are in `depth_results.zip`.

4.7. Efficiency

Table 6 lists the measurements. DAv2-S and VDA-S use the same input size, so the cost of temporal consistency is $1.7\times$ the time and $19\times$ the memory. The global scale is free at inference.

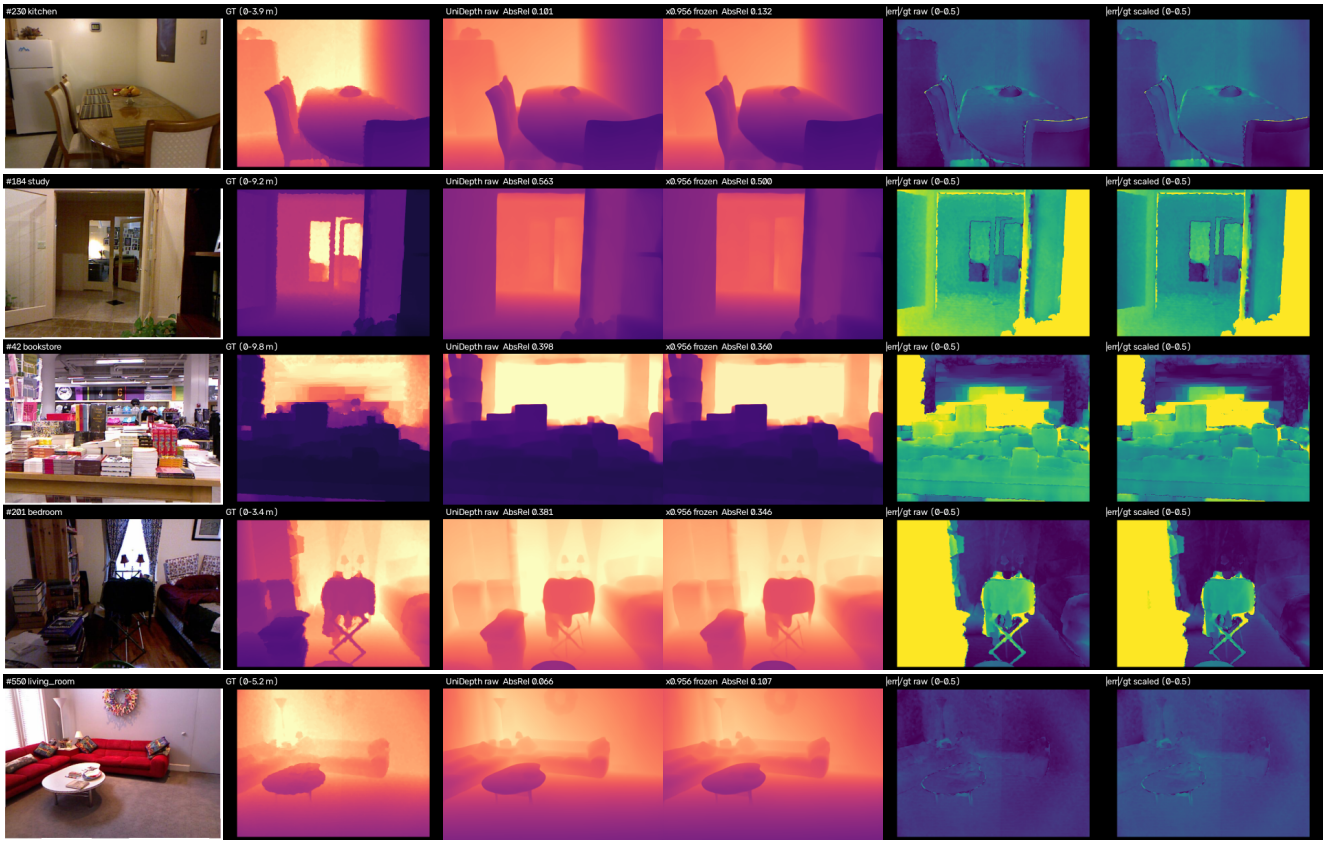


Figure 3. NYU test images. Columns: RGB, ground truth, raw UniDepthV2-S, frozen scale, and AbsRel error maps for both (0–0.5). Within a row all depth panels share one range. Row 1: the median-error image. Rows 2–4: the three highest-error images (study, bookstore, bedroom). Row 5: the image the scale hurts most.



Figure 4. `davis_rollercoaster`, frames 0, 35, and 69. Columns: RGB, DAv2-Small run per frame, and VDA-Small run on the clip. Both are relative depth (bright is near). Each method uses one fixed normalization over all 70 frames, so colors are comparable within a column but not across columns.

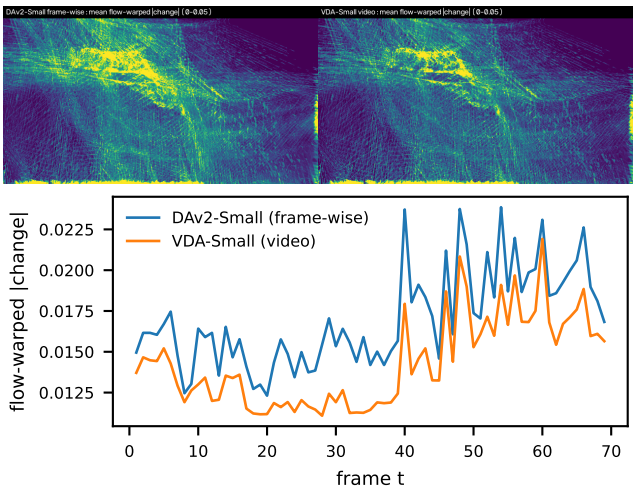


Figure 5. Optional temporal diagnostic. Top: per-pixel mean flow-warped change over the clip (black: flow rarely consistent). Bottom: per-frame mean change. Units are fractions of each method's fixed clip range.

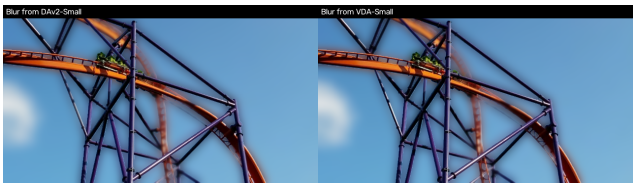


Figure 6. Depth-aware background blur on frame 35, driven by DAv2-Small (left) and VDA-Small (right), with the same focus level $\phi = 0.55$.

5. Conclusion

Stereo makes the geometry concrete: with calibration, 1.42 px of disparity error is millimeters near the camera and decimeters at the back of a room. RGB-only UniDepthV2-Small predicts NYU depth about 4% too far. One scale fitted on 795 training images, and frozen, removes part of that bias: AbsRel goes from 0.1145 to 0.1020, with RMSE and δ_1 improving too, in 13 of 16 scene types, and with a stable scale from about 50 images. Most of that bias comes from the predicted camera: with the true intrinsics the same correction is worth only 0.0014 AbsRel. On the rollercoaster clip, Video Depth Anything-Small changes less between frames than frame-wise Depth Anything V2-Small, at $1.7\times$ the time and $19\times$ the memory.

Limitations. The scale is specific to one dataset and one camera. It cannot fix images whose scale differs from the average (study, bookstore) or whose shape is wrong (bedroom), and it moves 37% of images the wrong way. The video comparison has no ground truth and covers one 70-frame clip.

Future work. A per-image cue, such as the EXIF focal length or a detected object of known size, could address the image-specific residual that the oracle exposes (AbsRel 0.102 versus 0.060). Fitting the scale per scene type, or in disparity space, would test whether part of that residual is shared within a domain.

References

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A. Earlier experiments (not part of the main comparison)

Before the final brief, we tried three additional things. They use a different protocol and are not compared with Table 2. **Aligned relative depth on NYU.** Depth Anything V2-Small with a per-image scale and shift fitted to *test* ground truth reaches AbsRel 0.132; median scaling alone gives 1.35. The course protocol forbids test-set alignment, so this is only evidence that relative depth needs both a scale and a shift. **Known-height calibration.** A detected person or car rescales an image with a 1.70 m or 1.59 m height prior [14]. Only 60 of 654 test images cast a vote, and the calibrated AbsRel rises to 0.273 (UniDepth with Kinect intrinsics). **Obstacle ranking and a 1.5 m warning.** On that calibrated depth, pairwise order accuracy over 9,127 obstacle pairs is 0.932, and the warning has precision 0.911, recall 0.693, and F1 0.788. Flow-guided fusion was also run on two self-recorded phone clips; the final brief does not require self-recorded video.